

# Course: Networked Control

Computer session 05/06: Delays and packet dropouts

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# Outline

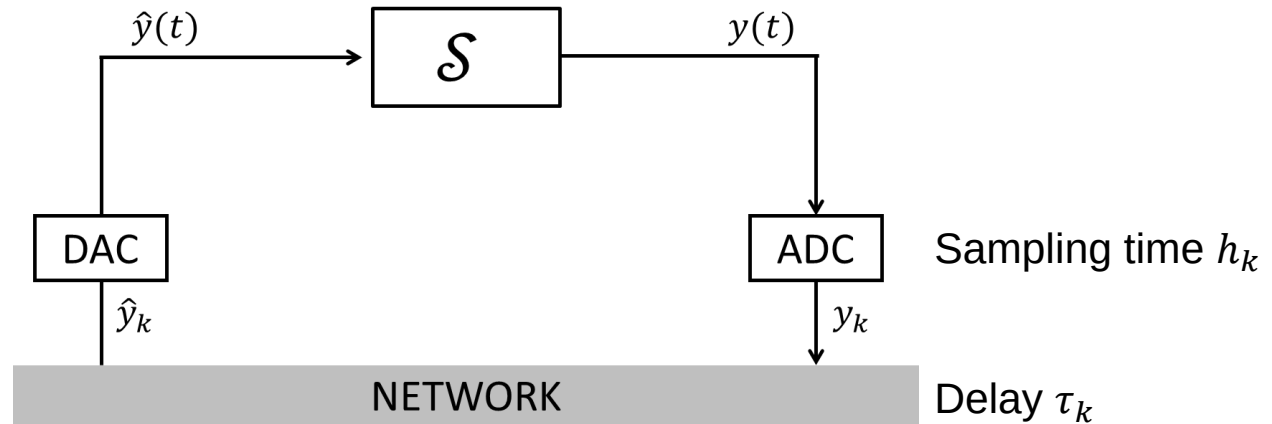
- NCS with delays and packet dropouts
- Exercise: Cart-stick balancer (inverted pendulum)
  - Analysis
  - Simulation

# Outline

- **NCS with delays and packet dropouts**
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# NCS with delays and packet dropouts

We consider the general NCS scheme



where the dynamics of  $\mathcal{S}$  is

$$\begin{cases} \dot{x}(t) &= Ax(t) + B\hat{y}(t) \\ y(t) &= Cx(t) \end{cases}$$

The dynamics of the NCS with delays is described by

$$z_{k+1} = \Phi(h_k, \tau_k)z_k$$

where  $z_k = [x_k^T, \hat{y}_{k-1}^T]^T$  and

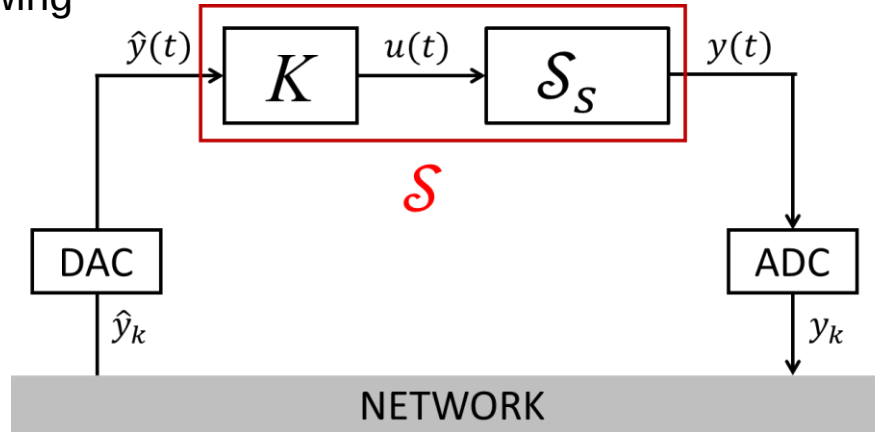
$$\Phi(h, \tau) = \begin{bmatrix} e^{Ah} + \Gamma(h - \tau)BC & e^{A(h-\tau)}\Gamma(\tau)B \\ C & 0 \end{bmatrix}, \Gamma(\xi) = \int_0^\xi e^{A\nu} d\nu$$

- When transmission takes place (**no dropouts**) the transition matrix is  $\tilde{\Phi}_1 = \Phi(h, \tau)$ .
- When there is a **dropout** the transition matrix is

$$\tilde{\Phi}_0 = \begin{bmatrix} e^{Ah} & \Gamma(h)B \\ 0 & I \end{bmatrix}$$

# NCS with delays and packet dropouts

In case system  $\mathcal{S}$  consists of system+output feedback controller co-located with the actuator, the scheme is the following



where the dynamics of  $\mathcal{S}_S$  is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e.,  $u(t) = K\hat{y}(t)$ . Renaming  $x(t) = x_S(t)$ ,  $A = A_S$ ,  $B = B_S K$ ,  $C = C_S$ , we retrieve the scheme of the previous slide. So, we rewrite

$$\tilde{\Phi}_1 = \Phi(h, \tau) = \begin{bmatrix} e^{A_S h} + \Gamma(h - \tau) B_S K C_S & e^{A_S(h-\tau)} \Gamma(\tau) B_S K \\ C_S & 0 \end{bmatrix}, \Gamma(\xi) = \int_0^\xi e^{A_S v} dv$$

$$\tilde{\Phi}_0 = \begin{bmatrix} e^{A_S h} & \Gamma(h) B_S K \\ 0 & I \end{bmatrix}$$

# NCS with delays and packet dropout

- 1) In case of **constant sampling time and delay**, i.e.,  $h_k = h, \tau_k = \tau$ , the NCS is exponentially stable **if and only if**  $\Phi(h, \tau)$  is Schur stable.
- 2) In case of **time-varying sampling time and delay**, assume that there exist constants  $h_{\min}, h_{\max}, \tau_{\min}, \tau_{\max}$  such that, for all  $k \in \mathbb{N}$ ,

$$\begin{aligned} 0 &\leq h_{\min} \leq h_k \leq h_{\max} \\ 0 &\leq \tau_{\min} \leq \tau_k \leq \tau_{\max} \end{aligned}$$

Then, the NCS is exponentially stable **if** there exists a symmetric matrix  $P > 0$  such that, for all  $h \in [h_{\min}, h_{\max}]$  and for all  $\tau \in [\tau_{\min}, \tau_{\max}]$

$$\Phi(h, \tau)^T P \Phi(h, \tau) - P < 0$$

- 3) In case of **deterministic dropouts** (and **constant sampling time and delay**)
  - a) The NCS is exponentially stable **if** there exists a symmetric matrix  $P > 0$  and scalars  $\alpha_0, \alpha_1$ , such that

$$\begin{aligned} \beta_0^r \beta_1^{1-r} &< 1 \quad (i) \\ \tilde{\Phi}_0^T P \tilde{\Phi}_0 &\leq \beta_0 P \quad (ii^*) \\ \tilde{\Phi}_1^T P \tilde{\Phi}_1 &\leq \beta_1 P \quad (iii^*) \end{aligned}$$

- b) Assume that there exists a symmetric matrix  $P > 0$  and scalars  $\beta_0 \geq 1, 0 < \beta_1 < 1$ , such that

$$\begin{aligned} \tilde{\Phi}_0^T P \tilde{\Phi}_0 &\leq \beta_0 P \quad (ii^*) \\ \tilde{\Phi}_1^T P \tilde{\Phi}_1 &\leq \beta_1 P \quad (iii^*) \end{aligned}$$

then, the NCS is exponentially stable for all  $r < \bar{r}$ , where

$$\bar{r} = \frac{1}{1 - \frac{\log \beta_0}{\log \beta_1}}$$

- 4) In case of **stochastic dropouts** (and **constant sampling time and delay**), the NCS, with dropout probability  $p$ , is exponentially mean-square stable **if** there exists symmetric matrix  $Z > 0$  such that

$$\begin{bmatrix} Z & \sqrt{p}(\tilde{\Phi}_0 Z)^T & \sqrt{1-p}(\tilde{\Phi}_1 Z)^T \\ \sqrt{p}(\tilde{\Phi}_0 Z) & Z & 0 \\ \sqrt{1-p}(\tilde{\Phi}_1 Z) & 0 & Z \end{bmatrix} > 0$$

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- NCS with delays and packet dropouts
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## Exercise: Cart-stick balancer (inverted pendulum)

Consider the cart-stick balancer system in the figure. Its model has three states, one input, and one output:

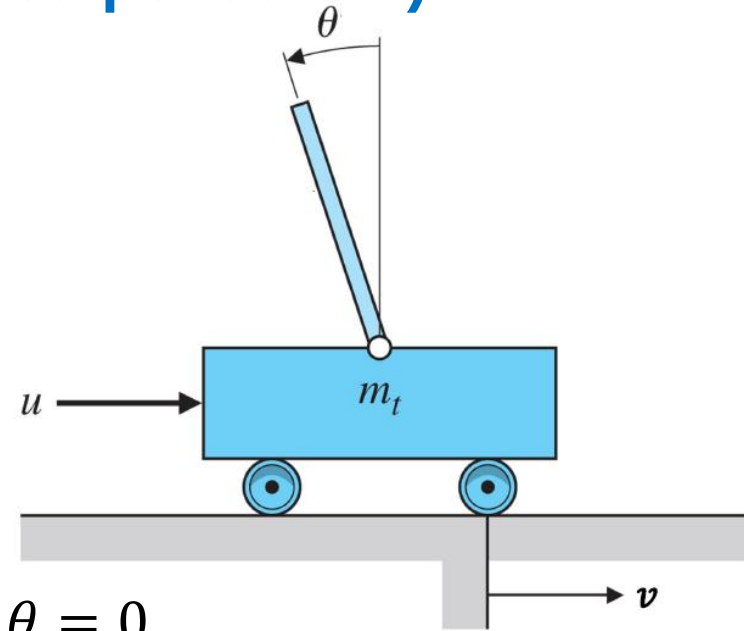
- $x_1: \theta/10$  – stick angle/10
- $x_2: \dot{\theta}/10$  – stick angular velocity/10
- $x_3 = v$  cart velocity
- $u$ : voltage of the motor driving the wheels
- $y = 70x_1 + 30x_2 + 0.1x_3$

Control goal: maintain the stick vertical, i.e., at  $\theta = 0$ .

The linearized model is

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 31.33 & 0 & 0.016 \\ -31.33 & 0 & -0.216 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ -0.649 \\ 8.649 \end{bmatrix} u$$

Assume that the system is controlled using a digital control law (i.e., with a sample-and-hold actuator). Consider different values of – constant – sampling time  $h$  (e.g.,  $h = 0.1, 0.01$ ).





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# Analysis

- Define the closed loop system matrix (in absence of delays)

$$F_{CL} = F + GKH$$

where

$$F = e^{As^h}, G = \Gamma(h)B_S, \Gamma(h) = A_S^{-1}(e^{As^h} - I)$$

for different values of the gain  $K$ . Find a range of values of  $K$  such that  $F_{CL}$  is Schur stable.

- For a selected value of  $K$  such that  $F_{CL}$  is Schur stable, in case of constant delay  $\tau \leq h$ , define the NCS transition matrix  $\Phi(h, \tau)$ . Compute the maximum value  $\tau_{\max}$  such that  $\Phi(h, \tau)$  is Schur stable.
- For the value of  $K$  selected at the previous point, in case of time-varying delay, compute the maximum value  $\tau'_{\max}$  of values that  $\tau_k$  can take, such that the NCS is guaranteed to be asymptotically stable. In practice, find  $\tau'_{\max}$  such that, for all  $\tau \in [0, \tau'_{\max}]$

$$\Phi(h, \tau)^T P \Phi(h, \tau) - P < 0$$

Hint: solve the problem with the “gridding” method, i.e.,

- Select  $\tau'_{\max}$
- Define a number  $N$  of values of  $\tau \in [0, \tau'_{\max}]$ , i.e.,  $\tau_1 = 0 < \tau_2 < \dots < \tau_N = \tau'_{\max}$
- Define an LMI feasibility problem as follows

```
yalmip clear
P=sdpvar(size(Phi));
L=[P>0];
for i=1:N
    L=L+[Phi(h,tau_i)^T P Phi(h,tau_i) - P < 0];
end
solvesdp(L)
```

- If the problem is not feasible reduce  $\tau'_{\max}$  and re-start from point 2.

# Analysis

- For the value of  $K$  selected at the previous points and for a selected value of the delay  $\tau$  such that, in case of constant delay,  $\Phi(h, \tau)$  is Schur stable, define matrices  $\tilde{\Phi}_1$  and  $\tilde{\Phi}_0$ . Define the value of (maximal dropout rate)  $\bar{r}$  according to Theorem 2.

Hint: solve the problem with the “gridding” method, i.e.,

1. Select  $\beta_0 \geq \rho(\tilde{\Phi}_0)^2$ ,  $\beta_1 \in [\rho(\tilde{\Phi}_1)^2, 1)$
2. Define an LMI feasibility problem as follows

```
yalmip clear
```

```
P=sdpvar(size(Phi));
```

```
L=[P>0]+[ $\tilde{\Phi}_0^T P \tilde{\Phi}_0 - \beta_0 P \leq 0$ ]+[ $\tilde{\Phi}_1^T P \tilde{\Phi}_1 - \beta_1 P \leq 0$ ]
```

```
solvesdp(L)
```

4. If the problem is feasible compute

$$\bar{r} = \frac{1}{1 - \frac{\log \beta_0}{\log \beta_1}}$$

5. Repeat 1.-4. for different values of  $\beta_0$ ,  $\beta_1$  and find the values that allow to maximize  $\bar{r}$ .

# Analysis

- For the value of  $K$  selected at the previous points and for a selected value of the delay  $\tau$  such that, in case of constant delay,  $\Phi(h, \tau)$  is Schur stable, define matrices  $\tilde{\Phi}_1$  and  $\tilde{\Phi}_0$ . Define the value of (maximal dropout probability)  $p$  according to Theorem 3. Hint: solve the problem with the “gridding” method, i.e.,

1. Select  $p = 1$

2. Define an LMI feasibility problem as follows

```
yalmip clear
```

```
Z=sdpvar(size(Phi));
```

$$L = \begin{bmatrix} Z & \sqrt{p}(\tilde{\Phi}_0 Z)^T & \sqrt{1-p}(\tilde{\Phi}_1 Z)^T \\ \sqrt{p}(\tilde{\Phi}_0 Z) & Z & 0 \\ \sqrt{1-p}(\tilde{\Phi}_1 Z) & 0 & Z \end{bmatrix} > 0 \quad + \quad [Z > 0]$$

```
solvesdp(L)
```

4. If the problem is not feasible reduce  $p$  and re-start from point 2.

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# Simulation

- For the selected value of  $K$ , simulate the trajectories of the closed-loop system in absence of delays

$$x_{k+1} = F_{CL}x_k$$

- For the selected value of  $K$ , simulate the trajectories of the NCS with constant delay  $\tau \leq \tau_{\max}$ :

$$z_{k+1} = \Phi(h, \tau)z_k$$

- For the selected value of  $K$ , simulate the trajectories of the NCS with time-varying delay

$$z_{k+1} = \Phi(h, \tau_k)z_k$$

where, for all  $k, \tau_k \leq \tau'_{\max}$

- Hint, to simulate a time-varying delay one can generate it randomly at each time step, e.g.  
`tauk=taumax*rand(1)`
- What happens if the maximum acceptable value of  $\tau_k$  is greater than  $\tau'_{\max}$ ? Recall that the condition of the theorem is only sufficient...
- For the selected value of  $K$  and with constant delay  $\tau \leq \tau_{\max}$ , simulate the trajectories of the NCS under packet losses with a rate  $r$ . Hints:
  - to simulate the system under scheduled losses of rate (e.g.) 6/10 we can, e.g.,

```
cont=1;
for k=1:kend
    if cont<=6
        Phi=Phi0;
    else
        Phi=Phi1;
    end
    cont=cont+1;
    if cont==11
        cont=1;
    end
    z(:,k+1)=Phi*z(:,k);
end
```

# Simulation

- For the selected value of  $K$  and with constant delay  $\tau \leq \tau_{\max}$ , simulate the trajectories of the NCS under packet losses with a rate  $r$ . Hints:
  - to simulate the system under random losses of probability  $p$  we can, e.g.,

```
for k=1:kend
    randnum=rand(1); % extracts the number for a uniform distribution in the
                    interval [0,1]
    if randnum<=p
        Phi=Phi0;
    else
        Phi=Phi1;
    end
    z(:,k+1)=Phi*z(:,k);
end
```
  - Make the simulations in case the rates are larger than the theoretical values found in the analysis. What happens?